INCH-POUND

MIL-A-53099(ME)
7 February 1990

MILITARY SPECIFICATION

ARCTIC FORWARD AREA REFUELING EQUIPMENT

(AFARE)

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an arctic fuel-dispensing system for use at all ambient temperatures from -60 to +95 °F in forward areas to refuel aircraft, ground vehicles, and other power equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1108	- Extinguisher, Fire, Vaporizing Liquid.
L-P-378	- Plastic, Sheet and Strip, Thin Gauge, Polyolefin.
P-D-680	- Dry Cleaning Solvent.
QQ-S-766	- Steel Plates, Sheets, and Strip - Corrosion Resisting.
QQ-S-781	- Strapping, Steel, and Seals.
VV-F-800	- Fuel Oil, Diesel.
WW-C-440	- Clamps, Hose (Low Pressure).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Preservation and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-C-3774	- Crates, Wood; Open, 12,000 and 16,000-Pound Capacity.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-27487	- Coupling Halves, Quick Disconnect, Cam-Locking Type.
MIL-F-46162	- Fuel, Diesel, Referee Grade.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-F-53091	- Filter-Separator, Liquid Fuel: Frame Mounted, 200 GPM Capacity, Arctic Service.
MIL-D-53092	- Drum, Fabric, Collapsible, Liquid Fuel, Cylindrical, 500-Gallon Capacity, Low Temperature.
MIL-H-53095	- Hose Assemblies, Rubber, Synthetic, Liquid Petroleum Fuels, Dispensing, Collapsible, Low Temperature.
MIL-H-53096	- Hoses and Hose Assemblies, Elastomer: Rubber, Liquid Fuel, Noncollapsible, Low Temperature.
MIL-T-83133	- Turbine Fuel Aviation, Kerosene, Type Grade JP-8.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Material.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are be those cited in the solicitation.

DRAWINGS

ME

TA13228E1800	- Arctic Forward Area Refueling Equipment.
13229E2400	- Frame Assembly, Extinguisher, 20 Pound, Halon.
13228E1768	- Gasket, Coupling Half, Quick Disconnect: Cam-Locking Type for Arctic Use.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-FS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 910 - Aviation Gasolines.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia (PA).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Arctic Forward Area Refueling Equipment, hereinafter referred to as "AFARE", shall be in accordance with TA13228E1800 and as specified herein. The major components of AFARE include the fuel drums, suction hose

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

assemblies, pump-engine assembly, discharge hose assemblies, pressure reducing assembly, filter-separator, and necessary hose connecting hardware and fittings to permit assembly of the equipment.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, interchangeability and operation of the AFARE. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Reliability. The specified mean-time-between-failure (MTBF) shall be 195 hours when the AFARE is operated at rated capacity and tested as specified in 4.5.2.2.

3.4 Maintenance ratio. The AFARE shall have a total maintenance ratio of not greater than 0.11 (see 6.6.2). A maintenance schedule shall be furnished prior to the start of any testing.

3.5 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.5.1 Material deterioration prevention and control. The AFARE shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.5.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.5.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with components and subcomponents and shall make information available, upon request, to the contracting officer or designated representative. Exterior finishes or treatments shall be approved for camouflage use by the contracting officer or designated representative (see 3.11).

3.5.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the AFARE may be newly fabricated from recovered materials to the maximum extent practicable,

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

provided the AFARE produced meets all other requirements of this specification. Used, rebuilt, or remanufactured components, pieces, and parts shall not be incorporated in the AFARE.

3.6 Components. The sizes, types, and quantities of components shall be as specified in parts list in TA13228E1800. The AFARE when assembled for field movement shall have all components as specified herein securely contained within the field packing and racking system.

3.7 Performance.

3.7.1 Capacity. The AFARE when assembled as specified herein for operation, and tested as specified in 4.5.2.4.1, shall deliver not less than 50 gallons per minute (gpm) simultaneously through each of the 2 nozzles with nozzle adapters.

3.7.2 Compatibility. The AFARE components shall be compatible when connected and shall not leak when tested as specified in 4.5.2.4.2. The AFARE components shall be suitable for use with hydrocarbon fuels conforming to MIL-T-5624, MIL-G-3056, ASTM D 910, MIL-T-83133, and VV-F-800.

3.7.3 Continuity. The static electric ground assemblies for the pump, filter-separator, and nozzles shall show a continuous electrical circuit when tested as specified in 4.5.2.4.3.

3.8 Human factors. The AFARE shall conform to the controls and ground workspace design requirements of MIL-STD-1472.

3.9 Safety and noise.

3.9.1 Safety. The AFARE shall present no uncontrolled hazards to the user. All shafts or rotating parts of the pumping unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Adequate shielding may be used to attain the same result. External surfaces shall be free of sharp edges and corners except where sharp edges and corners are necessary and are not detrimental to safety. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Requirement 1 of MIL-STD-454 shall apply.

3.9.2 Noise limits. The noise produced by the AFARE shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3 and 5.4, when tested in accordance with 4.5.2.6. The provisions of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3, except the sign will state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET." The sign shall be readable at the distance indicated on the sign.

3.10 Environmental requirements. The AFARE shall conform to the following environmental requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

3.10.1 Operating temperature. The AFARE shall perform as specified herein in all ambient temperatures between -60 and +95 °F.

3.10.2 Storage temperature. The AFARE shall not be damaged by storage in all ambient temperatures between -65 and +145 °F.

3.11 Treatment and painting. The portions of the AFARE normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383, conforming to MIL-C-46168.

3.12 Metallic component finish. Aluminum parts shall have a dyed, anodized coating conforming to MIL-A-8625, type II, class 2, color to approximate forest green.

3.13 Instructions. An instruction plate/placard shall be provided with a diagram showing the assembly of the AFARE components and a brief description of the assembly procedure. The instruction plate/placard shall be attached in a conspicuous location.

3.14 Workmanship. All parts and components of the AFARE shall be clean and free from extraneous material.

3.15 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed in the applicable QPL (see 6.4).

3.16 Cam-locking couplings. Cam-locking couplings shall conform to MIL-C-27487. Cam-locking couplings shall be anodized as specified in MIL-A-8625, type II, class I, thickness no less than 0.0004 inch for castings and no less than 0.0007 inch for wrought, and dyed to match color number 34094 of FED-STD-595. All coupling halves shall have vibration resistant cam-locks. The design shall be so that no pins are required to lock the arms in place.

3.17 Component field packing and racking system. A component field packing and racking system shall be provided which is reusable and contains and protects all hoses and components of the AFARE.

a. Two-man-portable metal containers or racks shall contain discharge hoses and small components and shall be designed to allow them to be positioned by rolling for short distances. The diameter of the cylindrical surface of the rack shall be appropriate for the minimum bend radius of coiled discharge hoses. Small components shall be contained in closable bags of low temperature waterproof material secured inside the rack in compartments to prevent movement when the container is rolled. Retaining straps or other means shall be provided

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

to secure the component bags and hoses inside the rack. Racks shall be stackable when containing hoses and components or when empty. Items to be contained shall include:

(1) Hoses
(2) Nozzles
(3) Fittings
(4) Ancillary equipment

b. Suction hose, which is less flexible than discharge hose, and ground rods shall be bundled with straps or other means into two-man-portable bundles.

c. The fueling manifold shall be mounted on a metal sled base with forklift pockets and shall have a removable protective cover of wood. The sled base shall have a towing ring on both ends and shall be towable on ice or snow in either direction.

d. The pressure reducing assembly shall be mounted on a metal sled base with forklift pockets and shall have a removable protective cover of wood. The sled base shall have a towing ring on both ends and shall be towable on ice or snow in either direction.

e. The exhaust gas heated air deflector and hose shall be protected in a wooden rack which shall be two man portable.

f. Portable rack for fire extingushers in accordance with 13229E2400.

3.18 Accessory items and bulk materials. Accessory items and bulk materials in the quantities shown on parts list on TA1322E1800 and in table I shall be supplied with each fuel system. The dust caps and plugs shall be attached to the assembly.

TABLE I. Accessory items and bulk materials.

Qty req'd	Nomenclature or description	Identification
1	Wrench, coupling, tank car, 1 3/4 width	5120-00-357-8431
2	Coupling, Rail Tanker, NATO	13222E8219
1	Seal, strapping, cres, type 201, 3/4 width, 100 EA per box	WW-C-440
1	Seal, strapping, cres, type 201, 1/2 width, 100 EA per box	WW-C-440
1	Strapping, cres, type 201, 3/4 width, 100 ft. per roll	
1	Strapping, cres, type 201, 1/2 width, 100 ft. per roll	
3	Extinguisher, fire, halon 1211, type F	A-A-1108
1	Frame assembly, extinguisher, 20 lb, Halon	13229E2400
1	Reducer, Quick disconnect, 2 in. male x 3 in. female	13228E1809
1	Reducer, quick disconnect, 3 in. male x 2 in. female	13228E1807
1	Reducer, quick disconnect, cam locking type, 1 in. female x 2 in. male, type XI, class 1	MIL-C-27487

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE I. Accessory items and bulk materials. (Cont'd)

Qty req'd	Nomenclature or description	Identification
1	Reducer, quick disconnect, cam locking type, 2 in. female x 1 in. male, type XI, class 1	MIL-C-27487
1	Coupling half, dust cap, 2-in.	MS27028-11
1	Coupling half, dust plug, 2-in.	MS27029-11
2	Gasket, quick disconnect coupling	13228E1768-6
1	Coupling half, dust cap, 2-in.	MS27028-5
1	Coupling half, dust plug, 2-in.	MS27029-5
2	Gasket, quick disconnect coupling	13228E1768-3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.6.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).
d. Inspection comparison (see 4.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.3 First article inspection. Unless otherwise specified, one first article AFARE system shall be examined and tested as specified in 4.5.1 and 4.5.2. Presence of one or more defects and failure of any test shall be cause for rejection. Unless otherwise specified, the Government reserves the right to perform first article testing on three or more AFARE systems. The tests specified in 4.5.2 shall be those which the Government desires to perform.

4.3.1 Rejection and retest. When an AFARE system fails to meet any of the requirements of the first article tests, the cause of failure shall be determined. Full particulars of the failure and the action required to correct the fault shall be reported to the contracting officer. All first article tests which are affected by the corrections shall be repeated.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.4. Failure of any test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. Each AFARE system shall be examined as specified herein for the following defects. Presence of one or more defects shall be cause for rejection.

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration in the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, finishes, or treatments.
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the AFARE system.
106. Any part (or component) not in accordance with the QAP requirements as shown on the drawings.
107. Any dimensions, other than those identified on the QAP requirements shown on the drawings, not as specified.
108. Components missing or not as specified.
109. Hardware not as specified.
110. Human factors engineering not as specified.
111. Treatment and painting not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

112. Metallic finish not as specified.
113. Instruction plate/placard not as specified.
114. Workmanship not as specified.
115. Lubrication procedure not as specified.
116. Component field packing and racking system not as specified.
117. Accessory items and bulk materials not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The AFARE shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by the maintenance schedule prepared and submitted by the contractor prior to test. Before performing the following tests, the AFARE shall be serviced with military fuels and lubricants designated for use at the ambient temperature at which the test is being conducted (see 6.4). The test fluid shall be gasoline conforming to MIL-G-3056 or ASTM D 910, jet fuel conforming to MIL-T-5624 or MIL-T-83133, diesel fuel conforming to MIL-F-46162 or VV-F-800, or solvent conforming to P-D-680.

4.5.2.2 Reliability. At least three but no more than five AFARE's shall be tested as specified in 4.5.2.4 for 650 hours with one system accumulating at least 200 hours. The lower test value shall be 65 hours and the upper test value shall be 195 hours. Acception shall be on 5 or fewer failures and rejection shall be on 6 or more failures (see 6.6.3). Nonconformance to 3.3 or 3.4 shall constitute failure of this test.

4.5.2.3 Maintainability.

4.5.2.3.1 Maintenance support evaluation. Maintenance operations required during testing shall be accomplished to determine conformance to 3.4. Failure of the support elements to maintain the AFARE as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.3.2 Maintenance ratio. The maintenance ratio shall be computed during testing. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.4 Performance.

4.5.2.4.1 Capacity. The AFARE shall be assembled for operation as specified in the applicable component technical manual. The AFARE shall be operated for a period of 48 continuous hours and flow rate shall be determined at the beginning of each hour by means of calibrated flowmeters. Nonconformance to 3.7.1 shall constitute failure of this test.

4.5.2.4.2 Compatibility. With the AFARE assembled and operating as specified in 4.5.2.4.1, an inspection of all components and fittings shall be made. Nonconformance to 3.7.2 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

4.5.2.4.3 Continuity. With the AFARE assembled and operating as specified in 4.5.2.4.1, an ohmmeter shall be used to determine electrical continuity of less than 500,000 ohms from the origin of the system to each of the two refueling nozzles. Nonconformance to 3.7.3 shall constitute failure of this test.

4.5.2.5 Safety. The AFARE shall be evaluated throughout testing as specified in 4.5.2. Nonconformance to 3.9.1 shall constitute failure of this test.

4.5.2.6 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.7 Environmental.

4.5.2.7.1 High temperature. The AFARE shall be tested as specified in MIL-STD-810, method 501.2. The maximum ambient storage temperature shall be +145 °F, and the maximum operating temperature shall be +95 °F. The AFARE shall be operated at rated capacity for a period of 24 hours. Nonconformance to 3.10.1 and 3.10.2 shall constitute failure of this test.

4.5.2.7.2 Low temperature. The AFARE shall be tested as specified in MIL-STD-810, method 502.2. The minimum ambient storage temperature shall be -65 °F. The minimum ambient operating temperature shall be -60 °F. The AFARE shall be operated at rated capacity for a period of 24 hours. Nonconformance to 3.10.1 and 3.10.2 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select AFARE at any time during the contract production period and subject these AFARE to the examination specified in 4.5.1 and tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on AFARE selected at random from those which have been accepted by the Government and will not include the previously inspected first article AFARE. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of any AFARE to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the AFARE and shall be cause for refusal by the Government to continue acceptance of production AFARE until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on AFARE produced and accepted under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all AFARE

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 First article pack inspections. The first article pack shall be examined for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, all components that will comprise a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.7.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

118.	Pump assembly not preserved in accordance with the referenced document as specified for levels A or C.
119.	Filter-separator and replacement filter elements not preserved in accordance with the referenced document as specified for levels A and C.
120.	Discharge hoses not drained and open ends not closed or covered as specified for levels A or C.
121.	Discharge hoses not coiled and secured in the rack as specified for levels A or C.
122.	Nozzles and fittings not placed in the rack and secured as specified for levels A or C.
123.	Suction hoses and ground rods not placed in the container and secured as specified for levels A or C.
124.	Consolidation not as specified for levels A or C.
125.	Packing not as specified for levels A or C.
126.	Marking missing, illegible, incorrect, or incomplete for levels A or C.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the packing requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.7). The first article pack may be accomplished utilizing either the first article model AFARE or a production AFARE. If the first article model AFARE is utilized, any preservation and packing shall be

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model AFARE and production AFARE.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Pump. Components of the pump shall be preserved in accordance with submethod IIA of MIL-P-116.

5.2.1.2 Filter-separator and replacement filter elements. The filter-separator and replacement filter elements shall be preserved in accordance with MIL-F-53091.

5.2.1.3 Discharge hose and fittings. Discharge hoses (MIL-H-53095) shall be drained of fuel or water and shall be coiled flat. Open ends shall be closed with appropriate caps or plugs or shall be covered with material conforming to L-P-378, type I, class 1, thickness 4 mil, secured in place with tape conforming to PPP-T-60, type IV. The coiled hoses shall be secured in a field packing and racking system (see 3.17). Nozzles and fittings shall be placed in the rack in their designated places and shall be secured.

5.2.1.4 Suction hose and ground rods. The suction hoses shall be preserved in accordance with the level A requirements of MIL-H-53096. Ground rods shall be bundled and wrapped with cushioning material to prevent damage to hose surfaces when hoses and rods are packed together.

5.2.1.5 Collapsible drums. The drums shall be preserved in accordance with level A requirements of MIL-D-53092.

5.2.1.6 Pressure reducing assembly and fueling manifold. The pressure reducing assembly and fueling manifold shall have open ends closed with appropriate caps or plugs which are attached to the units.

5.2.1.7 Gas cans with 5 feet of hose and fuel cap adapter. The gas cans with the hose and adapter shall be placed in a close-fitting box conforming to PPP-B-601, style I, domestic type. Blocking and bracing shall be in accordance with MIL-STD-1186 to prevent movement and damage.

5.2.1.8 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.9 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, level A.

5.2.1.10 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.11 Consolidation. The repair parts, maintenance tools, and technical publications for each AFARE preserved as specified shall be consolidated in a

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

close-fitting box conforming to PPP-B-601, overseas type, style I, grade B. Closure shall be in accordance with the box specification. Strapping is not required.

5.2.2 Level C.

5.2.2.1 Pump. Each pump shall be filled with the proper operating oil and covered with a waterproof barrier shroud. The shroud shall be secured at the base of the pump.

5.2.2.2 Filter-separators and replacement elements. The filter-separators and replacement elements shall be preserved as specified for level A.

5.2.2.3 Pressure reducing assembly and fueling manifold. The pressure reducing assembly and fueling manifold shall be preserved as specified for level A.

5.2.2.4 Collapsible drums. The collapsible drums shall be preserved in accordance with the level C requirements of MIL-D-53092.

5.2.2.5 Remaining components. The remaining components shall be preserved as specified for level A.

5.2.2.6 Consolidation. The repair parts, maintenance tools, and technical publications shall be consolidated as specified for level A.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Pump, pressure reducing assembly and fueling manifold. The pump, pressure reducing assembly and fueling manifold shall be packed in individual boxes conforming to PPP-B-601, overseas type, style a, modified as described herein. The containers shall be designed to be reusable by assembling the sides, end, and top as a reusable box cover. Battens, 2 by 2 inches in cross section, shall be positioned on the upper surface of the box base to provide a means to position the cover of the box properly and prevent lateral and longitudinal movement of the cover. Battens shall be located a distance from the edges of the box base equal to the thickness of the cleat and plywood of each side and end panel. The cover shall be secured to the container base with strapping conforming to QQ-S-781, class 1, type I or IV, finish B, size as applicable. Strapping shall encircle the box girthwise both laterally and longitudinally; those straps perpendicular to the skids shall be placed between the skids and the lower surface of the box bottom panel.

5.3.1.2 Suction hoses and ground rods. The suction hoses and ground rods shall be secured in two-man-portable bundles with reusable ties and placed in a box conforming to PPP-B-601, domestic type. Strapping shall be in accordance with QQ-S-781, class 1, type I or IV, finish B, size as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

5.3.1.3 Filter-separator and replacement filter elements. The filter-separator and replacement filter elements shall be packed in accordance with level A requirements of MIL-F-53091.

5.3.1.4 Collapsible drums. The drums shall be packed in accordance with level A requirements of MIL-D-53092.

5.3.1.5 Exhaust gas heated air deflector and hose. The exhaust gas heated air deflector and hose shall be placed in a box conforming to PPP-B-601, domestic type. Closure shall be in accordance with the box specification. Strapping is not required.

5.3.1.6 Consolidation. All of the components of the AFARE preserved and packed as specified in 5.2 and 5.3 shall be consolidated in a container conforming to MIL-C-104, type II, class 2, style a. The contents of the container shall be braced, blocked, and tied down in accordance with MIL-STD-1186 to prevent damage and movement.

5.3.2 Level C.

5.3.2.1 Pump, pressure reducing assembly and fueling manifold. Each pump, pressure reducing assembly and fueling manifold shall be packed as specified for level A except the boxes shall be domestic type.

5.3.2.2 Suction hoses and ground rods. The secured suction hoses shall be packed as specified for level A except the strapping shall be finish A.

5.3.2.3 Filter-separators and replacement filter elements. The filter-separators and replacement elements shall be packed in accordance with level C requirements of MIL-F-53091.

5.3.2.4 Collapsible drums. The drums shall be packed in accordance with level C requirements of MIL-D-53092.

5.3.2.5 Exhaust gas heated air deflector and hose. The exhaust gas heated air deflector and hose shall be packed as specified for level A.

5.3.2.6 Consolidation. The components of the AFARE shall be consolidated into one container conforming to MIL-C-3774, type II. The contents shall be blocked, braced, and tied down in accordance with MIL-STD-1186 to prevent damage and movement.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AFARE is intended for refueling helicopters in forward operational areas and for refueling military vehicles and powered equipment at

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:12Z
Check the source to verify that this is the current version before use.

low temperature (-60 °F) environment. The AFARE shall be used in all environmental conditions listed herein.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is not required for inspection and approval and number of units required (see 3.2).
d. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and where, and which examination and tests shall be conducted by the contractor (see 3.2).
e. Top coat color when other than as specified (see 3.11).
f. Time frame required for submission of first article pack (see 5.1).
g. Level of preservation and packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the AFARE should be an initial production model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Lubricants. MIL-STD-838 prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of nonstandard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113. The contracting officer should note that unless otherwise authorized by the U.S. Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.5 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

" The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Definition.

6.6.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAPs indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6.2 Maintenance ratio. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before- and after-operation checks are excluded.

6.6.3 Failure. A failure is defined as any malfunction which the operator cannot correct within 30 minutes by adjustment, repair, or replacement action; using controls, on-equipment tools, or parts; and which cause or may cause:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Serious damage to the AFARE system by continued operation.
c. Serious personnel safety hazards.

6.7 First article pack. Any changes or deviations of production pack from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the AFARE in accordance with this specification.

6.8 Design drawings. The contracting officer shall specify that complete design drawings be provided to fully describe the component field packing and racking system.

6.9 Subject term (key word) listing.

Aircraft refueling
Fuel-dispensing systems
Ground vehicle refueling
Powered equipment refueling
Refueling system

Custodian:
Army - ME

Preparing activity:
Army - ME

(Project 4930-A273)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:12Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-53099(ME)	90/02/07

3. DOCUMENT TITLE
Arctic Forward Area Refueling Equipment (AFARE)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

[redacted]

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717

c. ADDRESS (Include Zip Code)
US Army Belvoir RDE Center
ATTN: STRBE-TSE
Ft. Belvoir, VA 22060-5606

IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
Defense Quality and Standardization Office
5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466
Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.